

Functional Specification Document (FSD)

MediAssist AI – Enterprise Healthcare Agentic AI Copilot

1. Purpose

Develop an enterprise-grade Healthcare AI Copilot that enables healthcare personnel to interact with structured and unstructured healthcare data through a unified conversational interface.

The solution shall support:

- Enterprise RAG
- Multi-Agent AI
- Multimodal AI
- MCP-based PostgreSQL Integration
- Evaluation Framework
- Azure Deployment

2. User Roles

Role	Description
Doctor	Access patient summaries, reports and recommendations
Hospital Administrator	Compliance and operational intelligence
Front Desk Staff	Appointment management
Insurance Team	Billing and claims verification
Clinical Operations Team	Escalation workflows and SOP lookup

3. Functional Modules

Module 1 – User Authentication

FR-001

System shall allow users to login.

Inputs

- Username
- Password

Outputs

- Authenticated session

Acceptance Criteria

- Valid credentials allow access
- Invalid credentials show error

Module 2 – Conversational AI Assistant

FR-002

System shall provide a conversational chat interface.

Inputs

Natural language questions

Examples

What is the discharge process?

Show patient history for P1005

Summarize uploaded report

Outputs

Context-aware AI response

Acceptance Criteria

- Response generated within 15 seconds
- Sources displayed

Module 3 – Document Upload

FR-003

System shall allow users to upload healthcare documents.

Supported Formats

- PDF
- DOCX
- TXT

Maximum Size

50 MB

Outputs

Document stored for processing

Module 4 – Image Upload

FR-004

System shall allow upload of healthcare images.

Supported Formats

- JPG
- PNG
- JPEG

Use Cases

- Prescription Analysis
- Lab Report Analysis
- Medical Report Understanding

Module 5 – Data Ingestion Pipeline

FR-005

System shall ingest uploaded documents.

Workflow

Document
↓
Parsing
↓
Chunking
↓
Embedding
↓
Vector Indexing

Outputs

Indexed vector records

Acceptance Criteria

Documents searchable through RAG.

Module 6 – Enterprise RAG

FR-006

System shall retrieve relevant enterprise knowledge.

Retrieval Sources

- SOPs
- Policies
- Guidelines
- Operational Manuals

Capabilities

- Semantic Search
- Hybrid Search
- Metadata Filtering

Outputs

Relevant contextual passages

Acceptance Criteria

Top-k relevant chunks returned.

Module 7 – Planner Agent

FR-007

System shall decompose complex requests.

Example

User Request:

```
Summarize report,  
retrieve patient history,  
check appointments.
```

Planner Output

```
Task 1:  
Analyze uploaded report
```

```
Task 2:  
Retrieve patient history
```

```
Task 3:  
Retrieve appointments
```

```
Task 4:  
Generate recommendation
```

Module 8 – Multimodal Agent

FR-008

System shall analyze uploaded healthcare images.

Capabilities

- OCR
- Information Extraction
- Vision Reasoning

Outputs

- Medicines
- Diagnoses
- Observations
- Dates

Acceptance Criteria

Extracted information displayed.

Module 9 – MCP Server

FR-009

System shall provide MCP-based access to PostgreSQL.

Direct database access by AI agents is prohibited.

MCP Tools

```
get_patient_history()
```

```
get_appointments()
```

```
get_lab_results()
```

```
get_billing_summary()
```

Acceptance Criteria

All database access occurs through MCP.

Module 10 – MCP Agent

FR-010

System shall invoke MCP tools dynamically.

Example

User:

```
Show lab results for patient P1021
```

Expected Action

```
MCP Agent
```

```
↓
```

```
get_lab_results()
```

```
↓
```

```
PostgreSQL
```

Module 11 – PostgreSQL Retrieval

FR-011

System shall retrieve structured healthcare data.

Tables

- Patients
- Appointments
- Billing
- Lab Results

Output

Structured records returned to Reasoning Agent.

Module 12 – Reasoning Agent

FR-012

System shall synthesize information from multiple sources.

Inputs

- User Prompt
- RAG Context
- MCP Data
- Vision Output

Outputs

- Summary
- Recommendations
- Grounded Response

Acceptance Criteria

Response includes source attribution.

Module 13 – Evaluation Agent

FR-013

System shall validate generated responses.

Evaluation Areas

- Faithfulness
- Relevance
- Grounding
- Hallucination Detection

Outputs

Confidence Score

Grounding Score

Hallucination Risk

Module 14 – Session History

FR-014

System shall preserve conversation history.

Capabilities

- Previous messages
- Context continuation
- Multi-turn conversations

Module 15 – Evaluation Dashboard

FR-015

System shall provide observability dashboards.

Metrics

- Retrieval Scores
- Agent Execution Trace
- Latency
- Token Usage
- Hallucination Rate

Module 16 – Azure Deployment

FR-016

System shall deploy all components to Azure.

Components

- Streamlit
- FastAPI
- MCP Server
- LangGraph
- PostgreSQL Connector

Acceptance Criteria

Accessible through Azure endpoint.

Sample User Stories

Doctor

US-001

As a doctor, I want to upload a discharge summary and retrieve patient history so that I can generate follow-up recommendations.

Administrator

US-002

As an administrator, I want to search compliance policies so that I can answer audit-related questions.

Front Desk Staff

US-003

As front desk staff, I want to retrieve appointment information so that I can assist patients quickly.

Insurance Team

US-004

As an insurance executive, I want to verify billing and policy information so that I can process claims.

Non-Functional Requirements

ID	Requirement
NFR-001	Modular architecture
NFR-002	Dockerized deployment
NFR-003	API-first design
NFR-004	Logging and tracing
NFR-005	Agent observability
NFR-006	Config-driven architecture
NFR-007	Error handling and retry mechanisms
NFR-008	Scalable retrieval pipeline